Lemma F for the octahedron: opposite petals never overlap

working notes

September 2026 — draft

Latest audit correction, 9 September 2026. The conditional far-fan reduction is false as stated. An exactly checked convex octahedron has V_2 overlapping W_0, while all other 27 face pairs are disjoint. The petal enters through the cut radial boundary wu_0, not an interior hinge. Its source is not sharpest; a new H-specific proof remains possible. See `n6/HINGE_AUDIT.md`. The 24-class reduction and the one-nonconvex-patch whole-net proof are withdrawn: use 49 original classes, with 2 excluded and 47 open. The independent curvature-pair and central-symmetry proofs are unaffected.

Second continuation, 9 September 2026. The proposed strategy that neither outer petal apex enters Ω is now refuted by a strictly convex integer-coordinate octahedron satisfying both selection rules. Its entire selected net is nevertheless nonoverlapping. See the counterexample near the end of the Case B discussion. The base-cone partition remains valid; the joint-petal problem and Lemma L remain open.

Earlier audit correction, 9 September 2026. The unrestricted double-outward half-plane claim in the previous draft is false; an integer-coordinate counterexample is checked by rational intervals in `n6/audit_witness.py`. Lemma 5 below is now restricted to the angle range used in Case A. The new base-cone lemma below supplies an exhaustive partition, including the larger angles and equality boundaries: only $\Sigma_W < \pi$ remains after Case A and the base-cone exclusion. The later audit above supersedes the former far-fan reduction. Neither these notes nor the numerical experiments prove the full six-vertex theorem. See `n6/REVIEW.md` for the full audit.

Setting and conventions

We use the conventions of `HANDOFF.md`. P is a convex polytope of octahedron type with antipodal (non-adjacent) vertices v, w and equator $u_0, \dots, u_3$ (the neighbours of w, indices mod 4, in counter-clockwise order around w seen from outside). The faces are $W_i = wu_iu_{i+1}$ and $V_i = vu_iu_{i+1}$; κ_x is the curvature at x; ω_i, ν_i are the angles of W_i at w and of V_i at v. The net Z_k cuts $\mathrm{star}(v) \cup \{wu_k\}$: the fan $W_k, W_{k+1}, W_{k+2}, W'_{k-1}$ around w (angular gap κ_w at the slit) with the petal V_i glued to W_i along u_iu_{i+1}. Touching counts as non-overlapping.

Only nine face pairs of Z_k share no net vertex: the three local pairs at the slit, the two opposite-petal pairs (V_k, V_{k+2}), (V_{k+1}, V'_{k-1}), and four petal-vs-far-fan pairs. Lemma F is the statement that under the hypothesis

$$\textbf{(H)} \qquad \kappa_v \;\geq\; \kappa_x \quad \text{for every vertex } x \text{ of } P \tag{1}$$

none of the six far pairs overlaps, in every Z_k. (The handoff's hypothesis $\kappa_v \geq \max_i \kappa_{u_i}$ is weaker; the numerics below show that the weaker hypothesis is *not* enough for the argument of §3, although no counterexample to the weaker statement itself is known.)

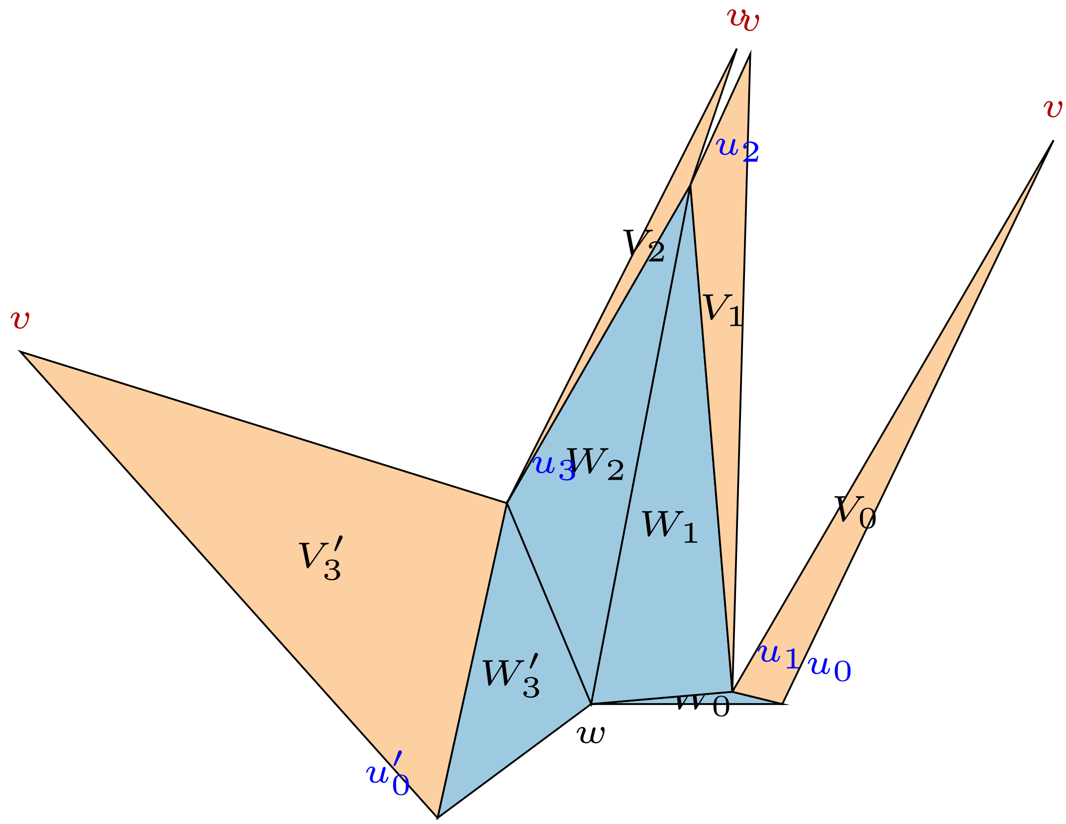

Figure 1: A typical net Z_k (fan faces blue, petals orange), v the sharpest vertex, slit at the sharpest neighbour u_0 of w.

Elementary consequences of (H). Since $\sum_x \kappa_x = 4\pi$ over six vertices, (H) gives $\kappa_v \geq 2\pi/3$, hence $\sum_i \nu_i = 2\pi - \kappa_v \leq 4\pi/3$. At any convex vertex each face angle is at most the sum of the other face angles (the spherical polygon inequality for the tangent cone), so $\nu_j \leq \frac{1}{2}\sum_i \nu_i$, i.e.

$$\nu_j \;\leq\; \pi - \tfrac{1}{2}\kappa_v \;\leq\; \tfrac{2\pi}{3} \qquad \text{for every } j. \tag{2}$$

1 Interior hinges and the unresolved far-fan pairs

Lemma 1 (petals do not cross interior hinges)**.** *Assume the three local pairs at the slit do not overlap. In Z_k no petal properly crosses the relative interior of a hinge segment wu_j, $j \neq k$.*

Proof. The two fan faces adjacent to wu_j are W_{j-1} and W_j; a petal crossing the segment would enter both. Away from the slit, a petal shares a net vertex with W_{j-1} or W_j, and the corresponding wedges are disjoint. There are two exceptions to this shared-copy argument: V'_{k-1} at wu_{k+1} and V_k at wu_{k-1}. In these cases a crossing would enter a local fan face, excluded by the hypothesis. $\square$

Remark 2 (Withdrawn far-fan implication). The previous draft claimed that local nonoverlap implies a petal meeting a far fan face also meets that face's petal. The first and last fan triangles each have a radial cut boundary. Lemma 1 does not exclude entry there, and the claimed implication is false.

An exact witness uses $v = 0, w = 1$ and $u_0 = 2, u_1 = 3, u_2 = 4, u_3 = 5$:

$$\begin{aligned} &v = (0, 0, 0), && w = (-89, -179, 150), && u_0 = (-595, 494, -131), \\ &u_1 = (295, -283, 90), && u_2 = (-7440, 5883, -1376), && u_3 = (-1166, 992, -271). \end{aligned}$$

Cut star(v) and wu_0. Only V_2 and W_0 overlap. The point $w + \frac{12}{25}(u_0 - w)$ in the developed cut edge lies strictly inside V_2. Strict supporting facets, overlap, and all 27 safe pairs have rational interval certificates. Vertex u_2 is sharper than v. A different original-edge tree gives a certified successful net.

Even after local and opposite-petal safety are established under H, the far-fan step needs an H-specific proof. The following Triple Lemma concerns the opposite petals only.

Conjecture 3 (Triple Lemma)**.** *Let W_i, W_j, W_{j+1} ($j = i + 1$) be three consecutive fan faces of a net and V_i, V_j, V_{j+1} their petals. Under (H), V_i and V_{j+1} do not overlap.*

Numerical evidence 4. No overlap of opposite petals under (H) in 72,000 random nets, nor under the weaker hypothesis $\kappa_v \geq \max \kappa_u$; adversarial hill-climbing (edges $\geq 0.05\,$diam, curvatures ≥ 0.05) brings the two petals to within 0.4% of the diameter but never into overlap. All observed overlaps (44 in 72,000 nets, without hypothesis) have two equator vertices sharper than v.

2 Where a meeting can be

Fix a triple as in Conjecture 3 and write $u = u_j$, $u' = u_{j+1}$ for the base of V_j, ϕ, ψ for the base angles of V_j at u, u', $\kappa = \kappa_u + \kappa_{u'}$. In the net, V_i and V_j meet at u only, separated there by the angular gap κ_u between the two copies uv_j and uv_i of the cut edge uv; likewise at u'. Let L be the line through uv_i and L' the line through $u'v_{j+1}$.

Lemma 5 (double-outward wedge, restricted angle range)**.** *Assume $0 < \phi + \kappa_u < \pi$ and $0 < \psi + \kappa_{u'} < \pi$. V_i lies in the closed half-plane of L not containing V_j, and V_{j+1} in the closed half-plane of L' not containing V_j. Hence every common point of V_i and V_{j+1} lies in $\mathcal{D} :=$ the intersection of these two half-planes, which is the wedge at the point $p^* = L \cap L'$ opposite to the base uu'. If $\kappa < \nu_j$ then $\mathcal{D}$ lies on v_j's side of the base line (above), and if $\kappa > \nu_j$ it lies on w's side (below); $\mathcal{D} = \emptyset$ if $\kappa = \nu_j$.*

Proof. The rays uv_i and $u'v_{j+1}$ make angles $\phi + \kappa_u$ and $\psi + \kappa_{u'}$ with the base, so they converge above the base iff $\phi + \psi + \kappa < \pi$, i.e. $\kappa < \nu_j$; the rest is the description of the four wedges at p^*. □

Remark 6. The range assumptions are automatic in Case A, since their sum is $\pi - \nu_j + \kappa < \pi$. They do not follow from (H) in general. Without them even the assertion that V_j is on the opposite side of L from V_i can fail. The rational counterexample in `n6/results/` refutes this unrestricted assertion, not the Case-A proposition.

Lemma 7 (the two cut edges never cross)**.** *In Case A ($\kappa < \nu_j$), the segments uv_i and $u'v_{j+1}$ are disjoint.*

Proof. If they crossed at p then p would be above the base with base angles larger than ϕ, ψ, so v_j lies inside the triangle $uu'p$ and $|up| + |u'p| > |uv_j| + |u'v_j| = |uv_i| + |u'v_{j+1}|$, contradicting $|up| \le |uv_i|$, $|u'p| \le |u'v_{j+1}|$. $\square$

Claim 8 (observed overlaps)**.** *In the 44 observed opposite-petal overlaps (no hypothesis), the intersection region lies in $\mathcal{D}$; 26 are above the base and 18 below, exactly as predicted by the sign of $\nu_j - \kappa$. In every below-base case the outer edges uu_i and $u'u_{j+2}$ of W_i, W_{j+1} converge on w's side (equivalently $\angle_u W_i + \angle_u W_j + \angle_{u'} W_j + \angle_{u'} W_{j+1} < \pi$) and exactly one of the two fan faces extends past the crossing point; both cannot, since fan faces are disjoint.*

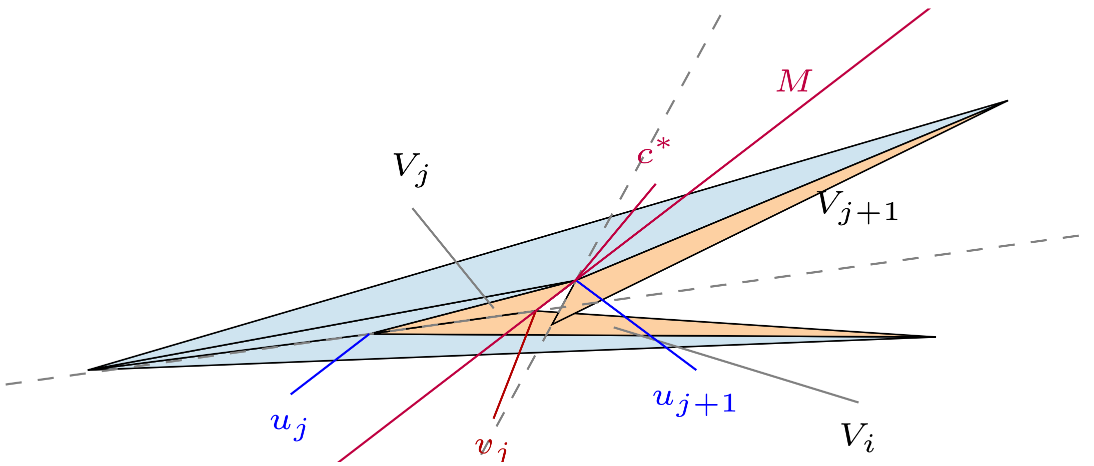

Figure 2: A thin above-base overlap (no hypothesis): the dashed cut-edge lines, the point c^* and the line M of §3. Here $\nu_i + \nu_j > \pi$ and V_i crosses M.

3 The rotation picture (above-base case)

Glue V_i, V_j, V_{j+1} flat around v with no gaps (the "v-fan"); this is possible because $\nu_i + \nu_j + \nu_{j+1} < 2\pi$, and in it the three petals occupy disjoint wedges at v. The net position of V_i is obtained from its v-fan position by the rotation ρ_u about u by the gap angle κ_u (away from V_j), and that of V_{j+1} by the rotation $\rho_{u'}$ about u' by $\kappa_{u'}$. Both rotations have the same sense. Therefore

$$V_i \cap V_{j+1} \neq \emptyset \text{ in the net} \iff \mathcal{R}(V_i^{\text{fan}}) \cap V_{j+1}^{\text{fan}} \neq \emptyset, \qquad \mathcal{R} := \rho_{u'}^{-1} \circ \rho_u,$$

and $\mathcal{R}$ is the rotation by $\kappa = \kappa_u + \kappa_{u'}$ about the point c^* on w's side of the base with $\angle c^* uu' = \kappa_u/2$, $\angle uu'c^* = \kappa_{u'}/2$ (composition of two rotations). Let M be the line c^*v; $\mathcal{R}$ moves points from u''s side of M towards u's side. Measure angles around c^* from M, positively towards u's side.

Lemma 9 (angular criterion)**.** *Let $\varepsilon_i \ge 0$ be the largest angle (seen from c^*) by which $V_i^{\text{fan}} \cap \{\text{above the base}\}$ crosses M towards u''s side, and ε_{j+1} the corresponding excursion of $V_{j+1}^{\text{fan}} \cap \{\text{above}\}$ towards u's side. If $\kappa < \nu_j$ and*

$$\varepsilon_i + \varepsilon_{j+1} \; < \; \kappa, \tag{3}$$

then V_i and V_{j+1} do not overlap in the net.

Proof. By Lemma 5 a common point p is above the base. The rotation ρ_u^{-1} maps the wedge of V_i at u in the net, $[\phi + \kappa_u,\ 2\pi - \angle_u W_j - \angle_u W_i]$, onto the v-fan wedge, so $x = \rho_u^{-1}(p)$ is below the base only if p is; hence $x \in V_i^{\mathrm{fan}}$ and $y = \rho_{u'}(p) \in V_{j+1}^{\mathrm{fan}}$ are above the base and $y = \mathcal{R}(x)$. Their angles from M satisfy $\theta_y = \theta_x + \kappa$ with $\theta_x \geq -\varepsilon_i$ and $\theta_y \leq \varepsilon_{j+1}$, so $\kappa \leq \varepsilon_i + \varepsilon_{j+1}$ (no wrap-around since $\kappa < \nu_j < \pi$). $\square$

In particular, if neither petal crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$), there is no overlap. When does a petal cross M? Write $\mu_u = \angle uvc^*$, $\mu_{u'} = \angle c^*vu'$, so $\mu_u + \mu_{u'} = \nu_j$ (c^* lies in the cone of V_j at v, being on w's side of the base between the lines vu, vu'). The wedge of V_i at v starts at vu and turns away from V_j by ν_i; it reaches the ray of M beyond v iff $\nu_i > \pi - \mu_u$. The other way for $V_i^{\mathrm{fan}} \cap \{\text{above}\}$ to cross M is through the far vertex u_i or through the point where the edge vu_i or uu_i crosses the base line.

Lemma 10. *Under (H) the two petals cannot both cross M through their wedges at v.*

Proof. That would need $\nu_i + \mu_u > \pi$ and $\nu_{j+1} + \mu_{u'} > \pi$, hence $\nu_i + \nu_{j+1} + \nu_j > 2\pi$, whereas $\nu_i + \nu_j + \nu_{j+1} \leq \sum \nu \leq 4\pi/3$. $\square$

Lemma 11 (the above-base case needs a very sharp v)**.** *Under (H), $\kappa < \nu_j$ forces $\kappa_v > 6\pi/7$, hence $\sum_i \nu_i < 8\pi/7$ and each $\nu_i < 4\pi/7$.*

Proof. $4\pi = \sum_x \kappa_x \leq 4\kappa_v + \kappa$ and $\kappa < \nu_j \leq \pi - \kappa_v/2$ by (2). $\square$

Numerical evidence 12. In 3000 random octahedra (all apexes, all triples with $\kappa < \nu_j$): the composition $\mathcal{R}$ has the stated centre and sense, and the criterion of Lemma 9 reproduces the net overlap test exactly. Under (H) neither petal ever crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$ in all 224 triples, degenerate ones included), and adversarial search under (H) cannot make either excursion positive. Under the weaker hypothesis $\kappa_v \geq \max \kappa_u$ a petal can cross M and (3) can fail (excursion 1.96 with $\kappa = 0.05$), so (H) is genuinely used. Every one of the 26 observed above-base overlaps violates $\varepsilon_i = \varepsilon_{j+1} = 0$ (20 through V_i, 6 through V_{j+1}).

Superseded: what remained for the rotation argument. Prove that under (H) and $\kappa < \nu_j$ neither $V_i^{\mathrm{fan}} \cap \{\text{above}\}$ nor $V_{j+1}^{\mathrm{fan}} \cap \{\text{above}\}$ crosses M (or, more weakly, that (3) holds). The three crossing mechanisms are: (a) the wedge at v: $\nu_i > \pi - \mu_u$; (b) the far vertex u_i above the base on u''s side of M; (c) an edge of V_i crossing the base line outside the segment uu' on u''s side of M. Mechanism (c) through vu_i requires $\nu_i + \nu_j > \pi + \psi$; (a) requires $\nu_i + \nu_j > \pi$; by Lemma 11 these need $\kappa_v < \pi$ and two of the four angles at v close to $4\pi/7$. The numerics say the fan faces then obstruct this, but the argument is not yet written.

4 Case A is closed: apex cones

The side conditions of §3 are not needed. Two lemmas finish the above-base case.

Lemma 13 (no wrap)**.** *Under (H), $\nu_i + \nu_j \leq \pi + \kappa_u + \kappa_{u'}$ and $\nu_{j+1} + \nu_j \leq \pi + \kappa_u + \kappa_{u'}$ for every triple.*

Proof. From $\sum_x \kappa_x = 4\pi$ and (H), $\kappa := \kappa_u + \kappa_{u'} = 4\pi - \kappa_v - \kappa_w - \kappa_{u_i} - \kappa_{u_m} \geq 4\pi - 4\kappa_v = 4\sum \nu - 4\pi$. If $\sum \nu \geq \pi$ then $\pi + \kappa \geq 4\sum \nu - 3\pi \geq \sum \nu \geq \nu_i + \nu_j$; if $\sum \nu < \pi$ then $\nu_i + \nu_j < \pi < \pi + \kappa$. $\square$

Lemma 14 (apex cones)**.** *Let C_i be the closed cone at v_i spanned by the cut edges v_iu and v_iu_i (it contains V_i), and C_{j+1} the cone at v_{j+1} spanned by $v_{j+1}u'$ and $v_{j+1}u'_m$. If $\kappa < \nu_j$, $\nu_i \leq \pi - \nu_j + \kappa$ and $\nu_{j+1} \leq \pi - \nu_j + \kappa$, then C_i and C_{j+1} have disjoint interiors. In particular V_i and V_{j+1} do not overlap.*

Proof. Measure directions counter-clockwise from $u \to u'$ and put $\alpha = \phi + \kappa_u$, $\beta = \psi + \kappa_{u'}$, so $\alpha + \beta = \pi - \nu_j + \kappa < \pi$. The cut edge uv_i has direction α (the gap turns v_j away from V_j) and V_i lies on its left, so from v_i the cone C_i is the sector $I_i = [\pi + \alpha - \nu_i,\ \pi + \alpha]$ ending at the direction e_1 towards u. Likewise $u'v_{j+1}$ has direction $\pi - \beta$, V_{j+1} lies on its right, and C_{j+1} is the sector $I_{j+1} = [-\beta,\ -\beta + \nu_{j+1}]$ starting at the direction towards u'. Two cones meet iff $d := v_{j+1} - v_i$ lies in the convex cone K generated by $I_i \cup (-I_{j+1})$, where $-I_{j+1} = [\pi - \beta,\ \pi - \beta + \nu_{j+1}]$. The two hypotheses say exactly that I_i and $-I_{j+1}$ lie inside $S = [\pi - \beta,\ \pi + \alpha]$, a sector of width $\alpha + \beta < \pi$ bounded by e_1 and by f_2 = the direction of $u' \to v_{j+1}$; hence $K = S$. Now $d = s_u e_1 + \ell e_0 + s_{u'} f_2$ with e_0 the base direction, and writing $e_0 = c_1 e_1 + c_2 f_2$ gives $c_1 = -\sin\beta / \sin(\alpha + \beta)$, $c_2 = -\sin\alpha / \sin(\alpha + \beta)$. Thus $d = (s_u - |up^*|)e_1 + (s_{u'} - |u'p^*|)f_2$ where p^* is the crossing of the two cut-edge lines, and $d \in S$ iff $s_u \geq |up^*|$ and $s_{u'} \geq |u'p^*|$, i.e. iff both cut edges reach p^*, i.e. iff they cross. Lemma 7 excludes this (equality means the cones touch at p^*). □

Proposition 15. *Under (H) the Triple Lemma holds whenever $\kappa_u + \kappa_{u'} < \nu_j$ (Case A).*

Proof. Lemma 13 gives the two angle conditions of Lemma 14. □

Numerical evidence 16. On 3241 random Case-A triples without hypothesis the cones never meet when the two angle conditions hold, and the conditions fail in 8% of them; under (H) they never fail (208 triples; adversarial margin 1.3 rad, `adv_wrap.py`), and the cone intersection has empty interior in every Case-A triple (adversarial depth -0.0007 diameters at the curvature guard, `adv_cones.py`). Without hypothesis the adversary makes the cones meet (depth $+0.006$), as it must, since 26 above-base overlaps exist.

Case B in this language. Any meeting lies in $C_i \cap C_{j+1}$, and in Case B this intersection can be non-empty. Cones at the two base vertices, rather than at the apexes, give the additional exclusion below.

5 An exhaustive partition: the base cones

Put $u = (0, 0)$, $u' = (\ell, 0)$, with the middle petal above the base and the middle fan face below it. Write

$$\alpha = \phi + \kappa_u, \quad \beta = \psi + \kappa_{u'}, \quad \sigma = \angle_u W_i + \angle_u W_j, \quad \tau = \angle_{u'} W_j + \angle_{u'} W_{j+1},$$

and $a = \angle_u V_i$, $b = \angle_{u'} V_{j+1}$. Vertex angle sums give

$$\alpha + \sigma + a = 2\pi, \qquad \beta + \tau + b = 2\pi, \qquad \alpha + \beta = \pi - \nu_j + \kappa.$$

All four angles $\alpha, \beta, \sigma, \tau$ are in $(0, 2\pi)$, and $a, b \in (0, \pi)$. The actual base cones of the outer petals have direction intervals $I = [\alpha, 2\pi - \sigma]$ at u and $J = [\pi + \tau, 3\pi - \beta]$ at u'. These are unwrapped intervals of widths $a, b < \pi$, valid even if α or β reaches or exceeds π.

Lemma 17 (base cones)**.** *If $\alpha + \beta \geq \pi$ and $\sigma + \tau \geq \pi$, the two base cones are disjoint. No curvature-ranking hypothesis is required.*

Proof. The left cone and the reversed right cone have their directions in $[\alpha, 2\pi - \sigma]$ and $[\tau, 2\pi - \beta]$, respectively. Both are contained in $S = [m, 2\pi - n]$, where $m = \min(\alpha, \tau) > 0$ and $n = \min(\sigma, \beta) > 0$. Since $a, b < \pi$,

$$m + n = \min(\alpha + \sigma, \alpha + \beta, \tau + \sigma, \tau + \beta) \geq \pi.$$

Thus S is a closed convex sector of width at most π, excluding the positive base direction. A meeting of the two translated cones would express $u' - u = (\ell, 0)$ as a vector in the left cone minus one in the right cone. That difference belongs to S, a contradiction. Equality is included. $\square$

Proposition 18 (complete partition). *Under (H), only $\Sigma_W := \sigma + \tau < \pi$ remains to prove for the opposite-petal statement. This regime automatically has $\kappa > \nu_j$. Neither $\kappa = \nu_j$ nor $\Sigma_W = \pi$ leaves a boundary case open.*

Proof. If $\kappa < \nu_j$, Case A applies. Otherwise $\alpha + \beta \geq \pi$, so Lemma 17 excludes a meeting when $\Sigma_W \geq \pi$. Finally $\alpha + \beta + \Sigma_W = 4\pi - a - b > 2\pi$, so $\Sigma_W < \pi$ implies $\alpha + \beta > \pi$, hence $\kappa > \nu_j$. When $\kappa = \nu_j$, the same identity gives $\Sigma_W > \pi$. $\square$

Remark 19. An exact integer-coordinate example in `n6/CASE_PARTITION.md` has (H), the sharpest-neighbour slit rule, and $\alpha > \pi$. Rational interval checks certify its curvature ordering and angular regime. Its fan-angle sum is above π, so Lemma 17 handles it. The unrestricted D-lemma remains false; it is not needed for the remaining outer-wedge argument.

Lemma 20 (remaining apex test). *In the remaining region $\Sigma_W = \sigma + \tau < \pi$, set $s = |vu|$, $s' = |vu'|$, and $\ell = |uu'|$. The left apex is strictly beyond the right outer fan-edge line, on the side away from w, if and only if*

$$\Sigma_W + a < \pi \quad \textit{and} \quad s \sin(\Sigma_W + a) > \ell \sin \tau.$$

The corresponding right-apex test is

$$\Sigma_W + b < \pi \quad \textit{and} \quad s' \sin(\Sigma_W + b) > \ell \sin \sigma.$$

Proof. The left apex is $s(\cos\alpha, \sin\alpha)$, while the right outer ray has direction $\pi + \tau$ from $(\ell, 0)$. The signed determinant, positive on the side away from w, is $s \sin(\tau - \alpha) - \ell \sin \tau$. The angle sums give $\alpha - \tau = 2\pi - \Sigma_W - a \in (0, 2\pi)$, so $\sin(\tau - \alpha) = \sin(\Sigma_W + a)$; positivity requires $\Sigma_W + a < \pi$. Reflection gives the right test. $\square$

Remark 21. The sine rule gives $s/\ell = \sin\psi / \sin\nu_j$ and $s'/\ell = \sin\phi / \sin\nu_j$. These are exact reformulations, not proofs of non-entry under the selection rules. The stronger candidate bounds $\Sigma_W + a \geq \pi$, $\Sigma_W + b \geq \pi$ have only numerical support. An exact abstract angle countermodel in `n6/CASE_PARTITION.md` shows that triangle sums, cone inequalities, and curvature rankings alone cannot prove these stronger bounds; metric realizability information is needed.

6 The below-base case (Case B)

By Proposition 18, it remains to consider $\Sigma_W < \pi$. This implies strict Case B and places both individual fan-angle sums below π. No restriction on $\phi + \kappa_u$ or $\psi + \kappa_{u'}$ is imposed here. For consistency with the historical formulas in this section, rename the two fan-angle sums α, β (they were σ, τ above). Write $u_k = u_{j+2}$ for the far vertex of W_{j+1} (in Z_k this is the slit copy, $k = i - 1$), A for the line through the outer edge uu_i of W_i, C for the line through $u'u_k$, $\alpha = \angle_u W_i + \angle_u W_j$, $\beta = \angle_{u'} W_j + \angle_{u'} W_{j+1}$ and $\Sigma_W = \alpha + \beta$ (the four fan angles at u, u'). Let $\kappa_{\text{back}} = \kappa_{u_i} + \kappa_{u_k} + \kappa_w$ be the total gap on the slit side, so that $\kappa_v + \kappa + \kappa_{\text{back}} = 4\pi$, and let $V_m = V_{i-1}$ be the fourth petal, glued to $W_m = W_k$. Throughout the remaining wedge analysis, $0 < \alpha, \beta$ and $\alpha + \beta = \Sigma_W < \pi$.

Lemma 22 (the wedge Ω). *V_i lies in the closed half-plane H_A of A not containing w, and V_{j+1} in the half-plane H_C of C not containing w; every common point lies in $\Omega := H_A \cap H_C$. The lines cross at a point X on w's side of the base, w lies in the triangle $uu'X$, and Ω is the wedge at X opposite to that triangle, of angle $\theta_X = \pi - \Sigma_W$.*

Proof. V_i and W_i lie on opposite sides of their common edge line A. On the base line, H_A is the ray beyond u away from u' and H_C the ray beyond u' away from u, so Ω misses the base line and lies on one side of it. The rays $u \to u_i$ and $u' \to u_k$ make the angles α, β with the base on w's side; they converge there since $\Sigma_W < \pi$, and X and Ω are below the base. w is on u''s side of A, on u's side of C and below the base. $\square$

Lemma 23 (how a petal reaches Ω). *Let $\Sigma_W < \pi$. V_i meets the interior of Ω iff u_i lies past X on A (*far vertex*, f) or $v_i \in \operatorname{int}\Omega$ (*apex*, a); likewise V_{j+1} with u_k and v_{j+1}. The far vertices u_i, u_k are never both past X.*

Proof. $V_i \subset H_A$ meets the open half-plane of C iff one of its vertices does; u is on w's side of C, and $u_i \in A$ is beyond C iff it is past X. If both far vertices were past X, then $X \in uu_i \cap u'u_k \subset W_i \cap W_{j+1} = \{w\}$, but $w \notin A$. $\square$

So a meeting needs $\Sigma_W < \pi$ and one of the sub-cases (a, a), (fa, a), (a, fa), (f, a), (a, f): at least one apex lies in Ω.

Lemma 24 (direction condition). *For every triple,*

$$\angle_{u_k} V_{j+1} - \Sigma_W = \nu_i + \nu_m + \angle_u V_i - \kappa_{\text{back}}, \qquad \angle_{u_i} V_i - \Sigma_W = \nu_{j+1} + \nu_m + \angle_{u'} V_{j+1} - \kappa_{\text{back}}.$$

If $\Sigma_W < \pi$, u_k is strictly before X, and v_{j+1} is in the interior of Ω, then $\angle_{u_k} V_{j+1} > \Sigma_W$. The mirror statement requires u_i strictly before X. No such direction conclusion is asserted here when the respective far vertex is already past X.

Proof. $\alpha = 2\pi - \kappa_u - \phi - \angle_u V_i$, $\beta = 2\pi - \kappa_{u'} - \psi - \angle_{u'} V_{j+1}$, $\phi + \psi = \pi - \nu_j$ and $\angle_{u_k} V_{j+1} = \pi - \angle_{u'} V_{j+1} - \nu_{j+1}$ give $\angle_{u_k} V_{j+1} - \Sigma_W = \kappa + \angle_u V_i - \nu_j - \nu_{j+1} - 2\pi$, and $\kappa = 2\pi + \sum \nu - \kappa_{\text{back}}$. For the direction: $u_k \in C$ lies on w's side of A and $v_{j+1} \in \Omega$, so the direction $u_k \to v_{j+1}$ points to the far side of C (the open half-circle of directions on V_{j+1}'s side of C) and across A; the direction of A towards u_i makes the angle Σ_W with $u_k \to u'$ (exterior angle at X), so the admissible directions are those making an angle in (Σ_W, π) with $u_k \to u'$, and $u_k \to v_{j+1}$ makes the angle $\angle_{u_k} V_{j+1}$. $\square$

Lemma 25 (sub-case (a,a): the hexagon identity). *Let $\Sigma_W < \pi$, neither far vertex past X, $v_i, v_{j+1} \in \Omega$, and assume the three local pairs at the slit do not overlap (Lemma L). If p is a common point of V_i and V_{j+1}, then*

$$\nu_m = \kappa_{\text{back}} + \gamma_p + \angle(p\, u_i\, v_i) + \angle(p\, u_k\, v_{j+1}), \qquad \gamma_p > 0,$$

where γ_p is the angle at p of the hexagon $u_i\, p\, u_k\, w\, u'_k\, v_m$. In particular a meeting needs $\nu_m > \kappa_{\text{back}}$.

Proof. The closed curve $\Gamma = u_i \to p \to u_k \to w \to u_i$ (segments in V_i, V_{j+1}, W_{j+1}, W_i) is simple: $[w, u_k]$ lies on w's side of A and meets $V_i \subset H_A$ at most at X, symmetrically for $[w, u_i]$. It lies in the triangle $\Delta = uu'p$, which is on w's side of the base line because $p \in \Omega$. Let Q be the region bounded by Γ. Its angle at w is $\omega_m + \kappa_w$ or $\omega_i + \omega_j + \omega_{j+1}$; in the second case W_j, hence V_j (which meets Γ nowhere, by the shared-vertex lemma), lie in $Q \subset \Delta$, but v_j is above the base. So Q contains W_m near w, hence W_m and V_m (their boundaries do not cross Γ: shared vertices with V_i, W_i, W_{j+1}, and Lemma L for the pairs with V_{j+1}, W_{j+1}). The hexagon $R = Q \setminus (W_m \cup V_m)$ has interior angles $\angle_{u_i} V_i - \angle(p\, u_i\, u) + \kappa_{u_i}$ at u_i, γ_p at p, $2\pi - \angle_{u_k} W_{j+1} - \angle(p\, u_k\, u')$ at u_k, κ_w at w, $2\pi - \angle_{u'_k} W_m - \angle_{u'_k} V_m$ at u'_k and $2\pi - \nu_m$ at v_m; their sum is 4π, and $2\pi - \angle_{u_k} W_{j+1} - \angle_{u'_k} W_m - \angle_{u'_k} V_m = \kappa_{u_k} + \angle_{u_k} V_{j+1}$. $\square$

Regime split for (a,a). If $\nu_m \le \kappa_{\text{back}}$ Lemma 25 excludes a meeting. If $\nu_m > \kappa_{\text{back}}$, the two outer cut edges $u_i v_i$ and $u_k v_{j+1}$ make the angles $\angle_{u_i} V_m + \kappa_{u_i}$ and $\angle_{u'_k} V_m + \kappa_{u_k} + \kappa_w$ with the segment $u_i u_k$, so their lines converge on V_m's side with angle $\nu_m - \kappa_{\text{back}}$, and Lemma 14 applies verbatim with the base uu' replaced by $u_i u_k$ (no-wrap $\nu_i, \nu_{j+1} \le \pi - \nu_m + \kappa_{\text{back}}$ follows from (H) as in Lemma 13): the cones C_i, C_{j+1} have disjoint interiors unless the two outer cut edges cross. Two gaps remain here: (i) a no-cross lemma for the outer cut edges across the slit ($u_k \ne u'_k$; the argument of Lemma 7 needs $|u_k v_m| \ge |u'_k v_m|$); (ii) the sub-cases with a far vertex past X.

Lemma 26 (the slit rule excludes the large-back-angle alternative)**.** *If u_k has maximum curvature among the four equator vertices, then for every face angle ν_t at v,*

$$\kappa_{u_k} + \kappa_w - \nu_t \;\ge\; \tfrac{1}{4}\kappa_v + \tfrac{3}{4}\kappa_w \;> 0.$$

In particular $\kappa_{\text{back}} > \nu_m$. Thus the (a,a) sub-case is excluded by Lemma 25, conditional on Lemma L. No sharpest-apex assumption is needed for this bound.

Proof. Gauss–Bonnet and the slit rule give $4\pi \le \kappa_v + 4\kappa_{u_k} + \kappa_w$, while the convex-vertex cone inequality gives $\nu_t \le \pi - \kappa_v/2$. Subtract these bounds, and use strict positivity of both curvatures. □

Numerical evidence 27. Under (H), in 23,578 Case-B triples, 19,858 have $\Sigma_W \ge \pi$ and only 27 have both petals reaching Ω (sub-cases (a,a) 22, (fa,a) 4, (a,fa) 1); no (f,f), as it must be. Seeded adversarial closest approach inside Ω (`caseB_sub.py adv`): (a,a) 3.5%, (fa,a) 3.2%, (a,fa) 1.9% of the diameter (the last slightly past the guard). Without hypothesis (a,fa) overlaps (+1.6%) and (a,a) stays 0.7% apart. In every (a,a) triple with $\nu_m > \kappa_{\text{back}}$ (109 found, with and without (H)) the apex cones have disjoint interiors, and the outer cut edges never cross when $\nu_m > \kappa_{\text{back}}$ (adversarial best -0.003 diameters, `adv_crossA.py`); $\nu_m > \kappa_{\text{back}}$ does occur under (H) in (a,a) (excess up to 0.30 rad), so both regimes are needed. An earlier count "15% of Case-B triples have both petals reaching the cone intersection" used a vacuous test for the position of X and is withdrawn.

Historical numerical evidence for the slit rule: superseded target. The old `adv_angle.py` mode `i` used the wrong base vertex; its saved margins remain withdrawn. The direct half-plane samples were a different experiment: 641 selected-rule Case-B triples suggested that neither outer apex enters Ω. This was numerical evidence only. The following exact counterexample refutes that stronger sufficient condition, so it is no longer a recommended route to the proof.

Exact counterexample to the individual-apex exclusion. Use

$$\begin{aligned} &v = (0,0,0), && w = (0,0,100), && u_0 = (81,0,65),\\ &u_1 = (-8,6,112), && u_2 = (-10,0,37), && u_3 = (9,-4,86). \end{aligned}$$

The equator is (u_0, u_1, u_2, u_3), and the cuts are the four edges at v and wu_0. Exact rational bounds verify the eight strict supporting faces, maximum curvature at v, and maximum equator curvature at u_0. For the triple W_0, W_1, W_2, they verify strict Case B, $\Sigma_W < \pi$, $\Sigma_W + a < \pi$, and $v_0 \in \operatorname{int}\Omega$. The far vertex u_0 is also past X. However, the entire other petal V_2 stays outside the left outer half-plane, so the petals do not overlap. All 28 pairs of the same net are independently certified nonoverlapping.

This is a failure of the individual-apex exclusion, not of the Triple Lemma or edge unfolding. Here the slit is at the first far vertex of the triple; the separate remote-apex bound in the opposite slit orientation must not be conflated with this assertion. The five reach sub-cases remain useful;

Lemma 26 still removes (a,a), conditional on Lemma L and the hypotheses of the hexagon lemma. The cases with a far vertex beyond X need joint control of the two petals.

The independent checkers are `n6/apex_entry.py` and `n6/polycert.py`, with the saved certificate `n6/results/caseB-apex-entry.certificate.json`. The interactive overview includes the figure and the exact six-vertex model.

What the intrinsic relaxations establish. The older rational angle model cannot close its spoke-length cycles. A newer metric model matches all twelve edge lengths, satisfies positive curvatures, all strict convex-vertex angle inequalities, and both strict rankings, yet still fails to close around the axis vw. A five-interval rational certificate excludes every possible value of $|vw|^2$ for that model. Thus local metric checks do not enforce original-facet realization. The closing condition is necessary, not asserted sufficient; the actual convex example above already shows that stronger realization tests cannot rescue the individual-apex claim. See `n6/AXIS_CLOSURE.md` for the equations and complete certificate.

7 Known false / do not retry

The individual-apex exclusion in Case B is false even under both selection rules, as shown above. Angular separation around w is also too strong for Lemma L: an exactly checked local pair shares rays but occupies different distances along them. For that example, a three-inequality radial calculation certifies a distance factor of at least 1217/1000 beyond the opposite fan edge. This is an example-specific separation, not a proof of Lemma L.

Simple separating lines for the two petals inside $\mathcal{D}$ (bisector of $\mathcal{D}$ at p^*, lines through v_j) fail in about a third of the non-degenerate cases even under (H): the separation is a rotation phenomenon, not a half-plane one. The six wedge–lean inequalities of the handoff fail in 0.5% of nets under (H) (needle-like w), so they cannot replace the argument above.

Code

`octa.py` (module: all quantities, nets Z_k, nine pairs, intersection polygons), `vwedge.py`, `oppcases.py` (collect/classify/draw overlaps), `adv_opp.py` (adversarial separation), `cstar.py` (checks of §3), `adv_eps.py`, `adv_mu.py`, `adv_cstar.py` (adversarial tests of (3) and of the crossing conditions), `tikz_net.py` (figures), `draw.py` (PNG); Case B: `caseB_sub.py` (sub-cases, seeded adversary), `caseB_draw.py`, `adv_crossA.py`, `adv_reach.py`, `adv_angle.py`, `adv_rule.py`, `lp_angles.py`.

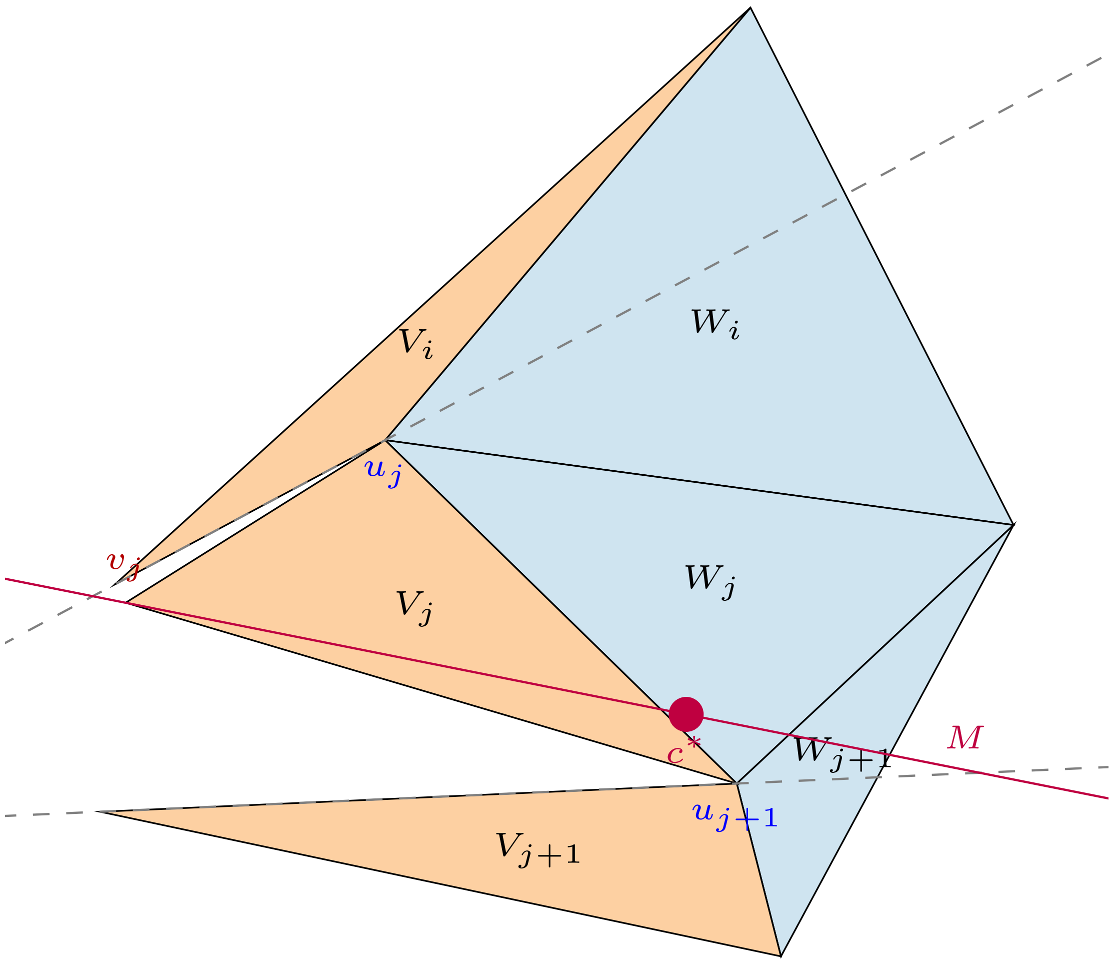

Figure 3: A triple under (H) with $\kappa < \nu_j$: c^*, the line M and the two dashed cut-edge lines; both petals stay on their own side of M.

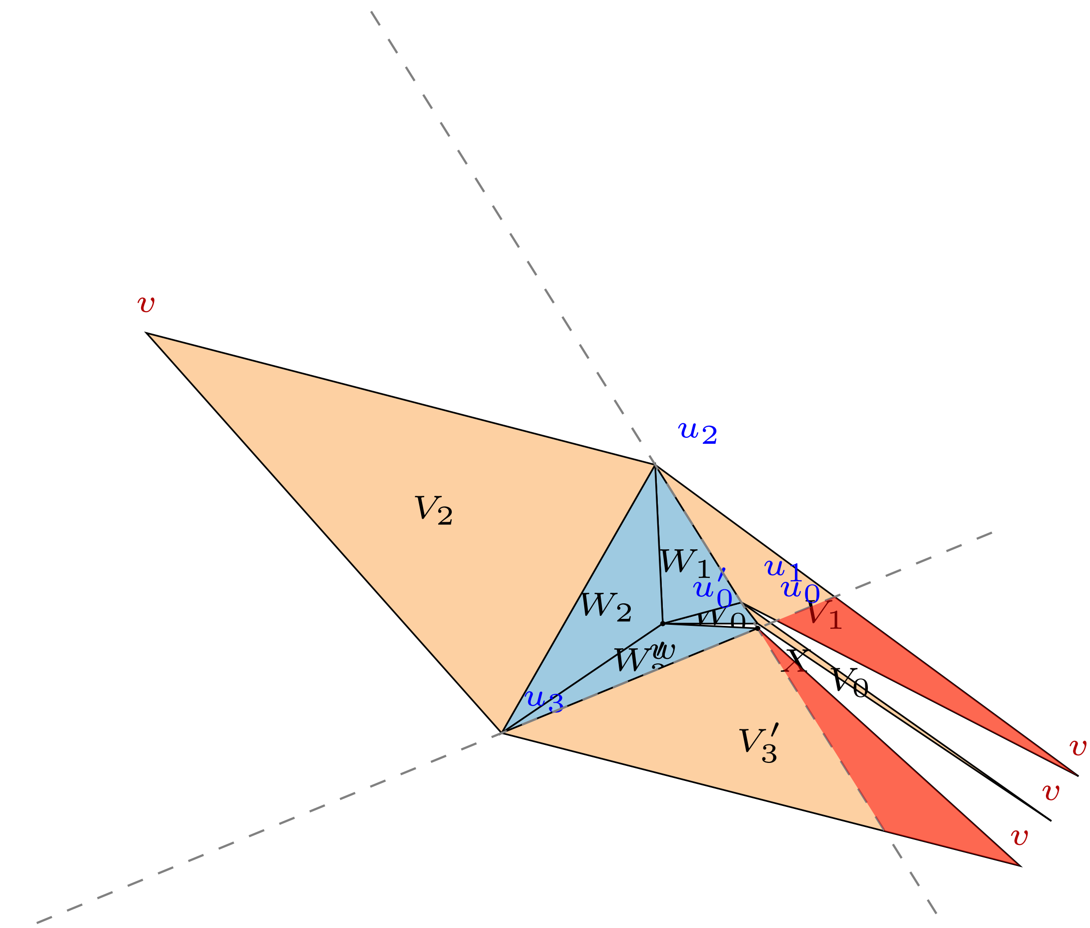

Figure 4: The tightest sub-case (a,a) configuration under (H) (`caseB_sub.py`): w, u_i, u_k nearly flat, u, u' needles, $\Sigma_W = 1.74$, the far vertex u_k at X. The two petal parts inside Ω (red) are separated by the thin fourth petal V_m; closest approach 3.5% of the diameter.